

Genome-wide identification and expression analysis of the mitogen-activated protein kinase gene family from banana suggest involvement of specific members in different stages of fruit ripening

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Abstract Mitogen-activated protein kinases (MAPKs) are important components of the tripartite mitogen-activated protein kinase signaling cascade and play an important role in plant growth and development. Although members of the MAPK gene family have been identified in model plants, little information is available regarding this gene family in fruit crops. In this study, we carried out a computational analysis using the *Musa* Genome database to identify members of the MAPK gene family in banana, an economically important crop and the most popular fruit worldwide. Our analysis identified 25 members of the MAP kinase (MAPK or MPK) gene family. Phylogenetic analyses of MPKs in *Arabidopsis*, *Oryza*, and *Populus* have classified these MPKs into four subgroups. The presence of conserved domains in the deduced amino acid sequences, phylogeny, and genomic organization strongly support their identity as members of the MPK gene family. Expression analysis during ethylene-induced banana fruit ripening suggests the involvement of several MPKs in the ethylene signal transduction pathway that are necessary for banana fruit ripening. Analysis of the *cis*-regulatory elements in the promoter regions and the involvement of the identified MPKs in various cellular processes, as analyzed using Pathway Studio, suggest a role for

the banana MPK gene family in diverse functions related to growth, development, and the stress response. This report is the first concerning the identification of members of a gene family and the elucidation of their role in various processes using the *Musa* Genome database.

Keywords Banana · Differential gene expression · Fruit ripening · Genome-wide analysis · MAP kinase

Introduction

The cellular machinery of all organisms includes a variety of intracellular signaling pathways that play important roles during developmental and environmental stresses. Mitogen-activated protein kinases (MAPKs) are serine/threonine kinases that have been demonstrated to participate in signal transduction through the MAPK signaling cascade in all eukaryotes (Hamel et al. 2006). In plants, the MAPKs are known to be involved in plant growth and development, programmed cell death, hormonal signaling, and the response to biotic and abiotic stresses (Hamel et al. 2006; Rao et al. 2010). Signaling cascades are organized into complex interconnected subcellular networks. The activation of MAPKs in response to changes in cellular status result in the transfer of signals to targets in different subcellular compartments to modulate various processes (Samajova et al. 2013). The MAPK signaling cascade, also known as the tripartite mitogen-activated protein kinase signaling cascade, primarily involves three types of genes, MAPK (MPKs), MAPKK (MKK), and MAPKKK (MKKK), which are linked to upstream and downstream regulators through a phospho-relay system. MPKs are activated when both tyrosine and threonine residues in the TXY/TDY motif are phosphorylated by dual-specificity kinases, the

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MPKKs, which in turn are activated by MPKKs via phosphorylation of the conserved Thr/Ser motif (Doczi et al. 2012; Chen et al. 2012). This MAPK module has been conserved during the evolution of higher plants (Janitza et al. 2012) and shows an unprecedented complexity. In the *Arabidopsis thaliana* genome, at least 20 MPKs, 10 MPKKs, and 60–80 MPKKs have been annotated so far (Chen et al. 2012), and a similar complexity has been reported in rice, *Medicago sativa*, *Zea mays*, tobacco, and tomato (Rao et al. 2010).

The MPK genes have been divided into subgroups depending on phylogeny, and each group has been assigned different functions (Janitza et al. 2012). Using *Arabidopsis*, the MPK genes and the robust nomenclature of MPKs from other species have been classified. Initially, it was thought that only the TDY type of MPKs exist in plants. However, recent studies related to the evolution of MPKs suggest that in algae, the MPKs have evolved as the TEY and TDY types, which were further distributed into different subgroups in land plants (Chen et al. 2012; Zhang et al. 2013). The TEY MPKs constitute group A–C, whereas the TDY MPKs are included in group D. The MPKs in group A are generally involved in developmental processes and abiotic and biotic stress responses, whereas those in group B are involved in pathogen-related responses and abiotic stresses (Chen et al. 2012; Zhang et al. 2013). The group C genes, for which limited information is available, are also involved in stress response (Zang et al. 2012). The groups D MPKs have a TDY activation domain and an extended C terminal.

Ethylene plays an important role in the ripening in climacteric fruits. Apart from being involved in various plant developmental and stress responses, the MPK pathway has also been shown to be involved in the ethylene signal transduction pathway (Cheong et al. 2003). However, their role in ethylene-regulated climacteric fruit ripening has not been studied in detail. Tomato is the only climacteric fruit for which limited information about the MPK gene family is available (Kong et al. 2012). However, the involvement of the tomato MPK gene family during ethylene-regulated fruit ripening has not been studied to date. In this study, we carried out a genome-wide identification of banana MPK genes using the complete genome sequence (D'Hont et al. 2012), and we classified the identified genes on the basis of their structure and function. We also studied the expression of different members of this gene family during ethylene-induced ripening. This study is the first in which a gene family has been characterized using the genome database from banana. Our analysis suggests that out of a total of 25 MPKs identified in the banana genome, a set of these are involved in ethylene-induced banana fruit ripening. We suggest that these MPKs might be involved in the ethylene signal transduction pathway leading to banana fruit ripening.

Materials and methods

In silico analysis to identify banana MPKs

To identify the MPKs in *Musa*, the whole genome sequence of *Musa acuminata* was downloaded from the database (<http://banana-genome.cirad.fr/>) (D'Hont et al. 2012). From the NCBI (<http://www.ncbi.nlm.nih.gov>) database, the sequences of 100 MAP kinases that belonged to more than 20 plant species were downloaded. A hidden Markov model (HMM) profile of these MAP kinases was generated using the Hmmer v 3.2 program (Finn et al. 2011), which was used as database to search similar proteins in the *M. acuminata* protein database. Using a highly stringent filter, 39 putative MPK genes were identified. Their nucleotide and genomic sequences were retrieved from the whole genome sequence of *M. acuminata* by the Coding DNA Sequence (CDSs) and gene files. To classify the exons and introns in the genomic sequences for each gene, the CDSs were mapped to the gene sequences using the Spidey program (Sayers et al. 2013) at NCBI. *Arabidopsis* MPK genes were downloaded from The *Arabidopsis* Information Resource (TAIR; <http://www.arabidopsis.org>; Tair10) (Lamesch et al. 2012), and the rice MPK genes were downloaded from The TIGR Rice Genome Annotation Resource (www.rice.plantbiology.msu.edu/) (Ouyang et al. 2007). The *Populus* MPK sequences were downloaded from the genome portal of the Department of Energy Joint Genome Institute (<http://genome.jgi.doe.gov>) (Grigoriev et al. 2012).

Alignment and phylogeny construction

The multiple sequence alignment of the MPKs from banana was carried out using the ClustalX program (Thompson et al. 1997). The domains were highlighted using the Boxshade program. For the phylogenetic analysis, MPKs from banana, *Arabidopsis*, *Oryza sativa*, and *Populus* were aligned using the ClustalX program, and the phylogenetic tree was constructed using the MEGA 5 program (Tamura et al. 2012). A maximum likelihood tree was constructed with the Jones, Taylor, and Thornton matrix with 100 bootstrap replicates. The chromosome localization for the banana MAPK genes was confirmed by BLASTX search using a local database containing the complete banana genome sequences of each chromosome. The identified MPKs were used in MCScanX (<http://chibba.pgml.uga.edu/mcscan2/>) software to investigate the evolutionary mechanism.

Plant material and treatments

Hands of mature unripe banana (*M. acuminata*; dwarf Cavendish, genome AAA, var. Robusta, *Harichhal*) were obtained from a local farm. This variety of banana does not ripen naturally and requires exposure of ethylene or other

hydrocarbons to initiate ripening for commercial purposes. Fruits from the same whorl of the hand representing similar developmental stages were treated with 100 $\mu\text{L/L}$ ethylene for 24 h at 22 °C in an air-tight container as described previously (Trivedi and Nath 2004). The containers were opened every 6 h, flushed with air to remove the accumulated CO_2 and maintain the O_2 concentrations, and replaced with 100 $\mu\text{L/L}$ fresh L ethylene each time. The fruits were allowed to ripen in air for 8 days under the same conditions and designated as day 0 (prior to ethylene treatment) to day 8 at the last stage of ethylene treatment. For 1-MCP treatment, the fruits were pre-exposed to 10 $\mu\text{L/L}$ of 1-methyl cyclopropane (Ethyl Bloc from Biotechnologies for Horticulture Inc., Walterboro, SC, USA) for 12 h followed by ethylene treatment as described above. Fruit pulp from the bananas was separated every 24 h for a period of 7 days after ethylene treatment, frozen in liquid nitrogen, and stored at -70°C . Peel of banana fruits after 4 days ethylene treatment was used to compare expression with pulp tissue of the same ripening stage.

RNA isolation and expression analysis

Total RNA from banana pulp was isolated according to Asif et al. (2006). First-strand complementary DNA (cDNA) was synthesized from total RNA using the Revert Aid H minus first-strand cDNA synthesis kit (Fermentas life Sciences, USA) according to the manufacturer's recommendations. The cDNA was analyzed by semiquantitative PCR, followed by agarose gel electrophoresis. The PCR mix for real-time PCR contained 1 μL of diluted cDNA (10 ng), 10 μL of 2 $\times$ SYBR Green PCR Master Mix (Applied Biosystems, USA), and 200 nM of each gene-specific primer (Supplementary Table S1) in a final volume of 20 μL . PCR with no template was also performed as the control for each primer pair. The expression of various MPKs was studied using the Applied Biosystems 7500 Fast Real-Time PCR System. All the PCRs were performed under following conditions: 20 s at 95 °C, 3 s at 95 °C, and 40 cycles of 30 s at 60 °C in 96-well optical reaction plates (Applied Biosystems, USA). The specificity of the amplicons was verified by melting curve analysis (60–95 °C) after 40 cycles. At least three biological replicates were analyzed for each cDNA.

Gene expression during fungus infection in banana roots

To study the modulation of the expression of banana MPK genes in response to fungus infection, we download the raw reads of the transcriptomes of the resistant mutant and susceptible Cavendish banana roots following inoculation with *Fusarium oxysporum* f. sp. All reads for the resistant mutant and susceptible Cavendish banana roots were mapped to MPK gene sequences using bowtie2 (<http://bowtie-bio.sourceforge.net/bowtie2/index.shtml>). More than one nucleotide mismatch was not allowed for the alignment of sequences.

The RPKM value (reads per kilo base per million mapped reads) was calculated for each MPK gene, and the absolute value of $\log_2$ ratio ≥ 1 was used as the threshold to assess the significant change in gene expression.

Identification of cis-regulatory elements and pathway analysis

To identify the *cis* regulatory elements, the upstream regions (1.5 kbp) of all banana MPK genes were extracted through a custom Perl script. This region was considered as the proximal promoter regions for the individual banana MPK genes. All the promoter sequences were placed in the Plant Care database (Lescot et al. 2002) in which a brief description of all motifs is presented. The role of MPK genes in different cellular processes was identified using the Pathway Studio software (Nikitin et al. 2003). For this, all of the MPK genes were subjected to blast against the TAIR database. The TAIR IDs of the aligned MPK genes were taken for cellular pathway analysis using the Pathway Studio software (Ariadne Genomics, USA) as described by Chakrabarty et al. (2009).

Results

MaMPK gene family in banana

A total of 100 MAP kinase genes from different plant species were downloaded from the GenBank database and used for the construction of the Hmmer profile (HMMER v3.2). The developed Hmmer profile was used for searching the *M. acuminata* protein database for putative MAP kinases. Our in silico search resulted in the identification of 1,139 sequences that had similarity to the Hmmer profile (Supplementary Information S2). After filtering on the basis of the *e* value ($<10^{-10}$), bias, and domains, 39 sequences were identified as putative MPKs in the banana. Furthermore, removal of redundancy resulted in 25 sequences that had similarity to the MPK according to the criteria used in the study. Out of these 25 sequences, three MPKs (*MaMPK-X1*, *MaMPK-X2*, and *MaMPK-X3*) are truncated. Out of these three, two (*MaMPK-X1* and *MaMPK-X2*) contain an activation domain. The genomic sequences and the coding sequences were downloaded from the *M. acuminata* genome and CDS database, respectively. Analysis of the genomic sequences of the truncated MPKs showed similar gene sequences, which revealed that these MPKs occur as truly truncated, even in the genomic sequence database. The average nucleotide length, encoded polypeptide length, and molecular weight of the banana MPKs are 1,219.6 bp, 405.64 amino acid residues, and 46.94 kDa, respectively. All the MPKs identified with their gene names, gene lengths, open reading frame (ORF) lengths, protein lengths, and other related information are provided in Table 1.

Table 1 Structural features of identified MaMPKs

Name	Chromosome	Position in chromosome	ORF size (bp)	Protein length	Predicted PI	MW (kDa)	Subcellular localization	MAPK Group	Gene_id
<i>MaMPK3-1</i>	6	32266722–32269510	1,104	367	5.584	43.17770414	Cytoskeleton	TEY	GSMUA_Achr6T33100_001
<i>MaMPK3-2</i>	9	2866043–2868202	1,110	369	5.95	43.30181804	Cytoskeleton	TEY	GSMUA_Achr9T04280_001
<i>MaMPK4</i>	11	2108460–2113196	1,101	366	6.527	42.74391674	Cytoplasm	TEY	GSMUA_Achr11T02910_001
<i>MaMPK7</i>	4	8092688–8098019	630	209	5.112	24.82989224	Mitochondria	TEY	GSMUA_Achr4T11270_001
<i>MaMPK6-1</i>	10	1660317–1665089	1,149	382	5.814	44.43891684	Cytoplasm	TEY	GSMUA_Achr10T00980_001
<i>MaMPK6-2</i>	5	4877126–4884041	1,149	382	5.711	44.41686454	Nuclear	TEY	GSMUA_Achr5T06640_001
<i>MaMPK6-3</i>	2	19424811–19433392	1,140	379	5.814	44.16354834	Nuclear	TEY	GSMUA_Achr2T19320_001
<i>MaMPK9-1</i>	1	15632693–15638527	1,803	600	6.517	69.18800594	Chloroplast	TDY	GSMUA_Achr1T20760_001
<i>MaMPK9-2</i>	9	557791–565882	1,836	611	6.92	70.17168374	Mitochondria	TDY	GSMUA_Achr9T00750_001
<i>MaMPK9-3</i>	4	2998593–3005565	1,593	530	8.233	60.99391644	Cytoplasm	TDY	GSMUA_Achr4T03750_001
<i>MaMPK9-4</i>	4	8067126–8074839	1,308	435	8.857	50.26024634	Cytoplasm	TDY	GSMUA_Achr4T11240_001
<i>MaMPK11-1</i>	4	3532382–3541077	1,281	426	6.3	49.45087464	Cytoplasm	TEY	GSMUA_Achr4T04470_001
<i>MaMPK11-2</i>	4	8400325–8406413	1,131	376	6.503	43.62282594	Cytoskeleton	TEY	GSMUA_Achr4T11650_001
<i>MaMPK11-3</i>	4	4290129–4296220	1,122	373	6.261	43.23248134	Cytoskeleton	TEY	GSMUA_Achr4T05570_001
<i>MaMPK11-4</i>	4	8987260–8996026	1,128	375	6.146	43.52077214	Cytoskeleton	TEY	GSMUA_Achr4T12170_001
<i>MaMPK14</i>	4	3008475–3013372	624	207	5.102	24.18203064	Cytoplasm	TEY	GSMUA_Achr4T03760_001
<i>MaMPK20-1</i>	9	4301242–4306507	1,764	587	9.377	66.99929374	Nuclear	TDY	GSMUA_Achr9T06770_001
<i>MaMPK20-2</i>	6	27689428–27693487	1,830	609	9.113	69.31038344	Nuclear	TDY	GSMUA_Achr6T27100_001
<i>MaMPK20-3</i>	random	84844349–84851035	1,875	624	10.204	71.37014394	Nuclear	TDY	GSMUA_AchrUn_randomT17900_001
<i>MaMPK20-4</i>	1	9111909–9118859	1,857	618	9.568	70.60876724	Cytoplasm	TDY	GSMUA_Achr1T11960_001
<i>MaMPK20-5</i>	11	11394313–11401540	1,749	582	9.447	66.89932324	Nuclear	TDY	GSMUA_Achr1T12400_001
<i>MaMPK20-6</i>	3	2039320–2046868	1,290	429	9.324	49.85229904	Cytoplasm	TDY	GSMUA_Achr3T03110_001
<i>MaMPKX-1</i>	9	27176481–27179877	309	102	5.831	12.40629044	Cytoplasm	TEY	GSMUA_Achr9T22050_001
<i>MaMPKX-2</i>	9	27176140–27176430	291	96	9.998	11.48862604	Cytoplasm		GSMUA_Achr9T22040_001
<i>MaMPKX-3</i>	11	2050959–2053213	320	107	5.848	12.94307774	Cytoplasm		GSMUA_Achr1T02860_001

Sequence alignment and phylogenetic analysis

As *MaMPK-X1*, *MaMPK-X2*, and *-MaMPK-X3* are truncated, these were not included in the alignment analysis. The alignment of the other 22 MaMPKs was carried out using the ClustalX program. The alignment clearly shows (Fig. 1) the presence of all the 11 domains in the MaMPKs, which are known to be present in MPKs from other plant species. The activation loop containing the TXY domain, which is phosphorylated for the activity, is present in all banana MPKs with a clear demarcation of sequences containing the TDY and TEY domains (Fig. 1). Interestingly, no banana MPK was identified with the TEY (MEY) domain. Previous report suggests that rice and *Solanum* species contain one MPK each with a MEY domain (Kong et al. 2012). The CD domain in the TEY type of MPKs is also present in the banana MPKs. The TDY domain-containing MPKs also had an extended C-terminal region, as is present in the TDY class of MPKs from other plants (Ichimura et al. 2002; Kong et al. 2012).

For the phylogenetic analysis, all 22 full-length MPK identified in this study, as well as 17 from *O. sativa*, 20 from *Arabidopsis thaliana* and 21 from *Populus trichocarpa*, were used. The phylogenetic analysis was performed using the MEGA5 program, and the ML tree was constructed with a bootstrap value of 100 using human Extracellular signal-regulated kinase (HsERK1) as an out-group. All the MPK clustered into four groups, which were designated as A–D (Fig. 2). Nomenclature of each banana MPK was carried out according to proximity with *Arabidopsis* MPKs present in different clusters (Hamel et al. 2006). According to the phylogenetic tree, many of the MPK genes in banana have paralogs among themselves; however, none of these paralogs were found as duplicons. The occurrence of paralogs in *Musa* was higher in comparison to any of the other species studied. However, for many of the MPK genes (at least 6), there were no orthologs present in banana.

Chromosomal distribution and gene structure of the MaMPKs

Out of the 25 MPKs in *M. acuminata*, 24 were successfully mapped to the 11 chromosomes; however one MPK

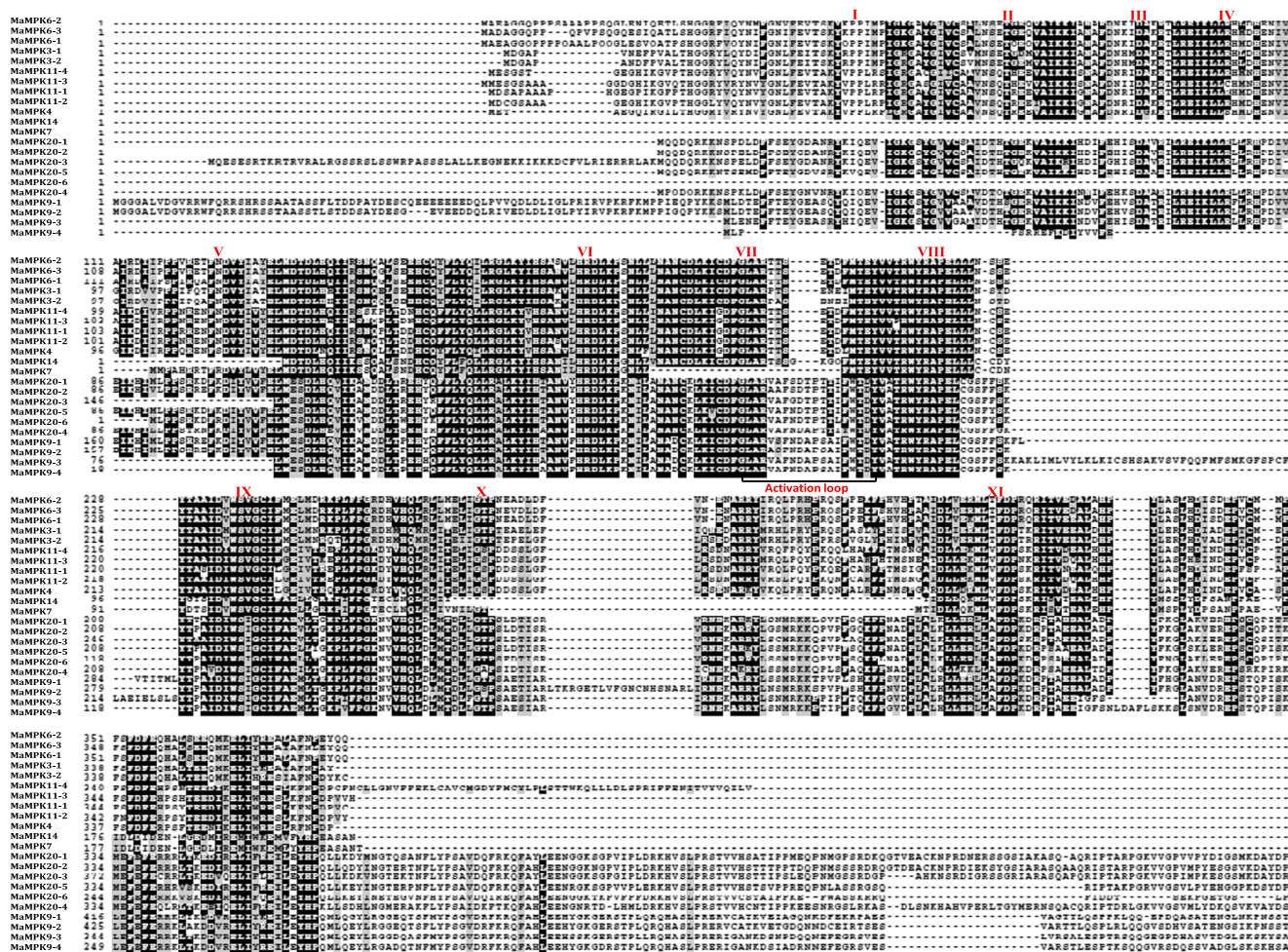
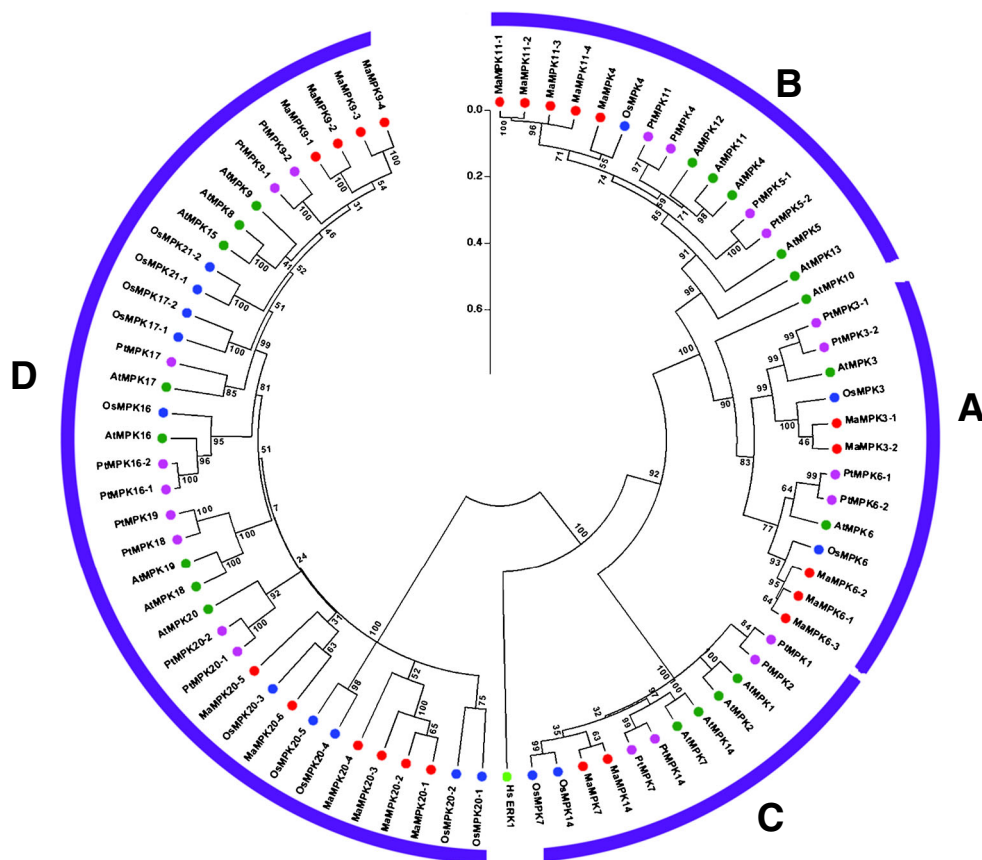


Fig. 1 Multiple sequence alignment of the MaMPK amino acid sequences. The alignment was performed using ClustalX. The conserved regions containing eleven domains (I–XI) present in banana serine/threonine protein kinases are shown in *Roman numerals*

Fig. 2 Phylogenetic analysis of the MaMPK gene family in banana. A maximum likelihood based phylogenetic tree of the banana, *Arabidopsis*, rice, and poplar MPK family genes was generated using the MEGA5 program. **A–D** indicate different gene clusters



(*MaMPK20-3*) was present in an uncharacterized chromosomal region (Fig. 3). Although there is high similarity among some of the MPKs, no duplication of the genes was observed. Most of the chromosomes to which the genes were mapped had two or more genes, except for chromosomes 2, 3, and 5, which mapped with only one MPK gene. No MPK gene was mapped on chromosomes 7 or 8. Interestingly, closely mapped MaMPKs on each chromosome did not show any clustering in the phylogenetic analysis (Fig. 2), and vice versa. None of the MPKs were found to be a tandem or segmental duplicated. There was no major clustering of the MPK genes, ruling out any possibility of tandem duplications. It seems that tandem duplication event is not playing an important role in MPK gene family expansion.

The gene structures of the 22 MPK genes were studied by aligning the genomic and CDS region for each MaMPK gene. As *MaMPKX-1*, *MaMPKX-2*, and *MaMPKX-3* were truncated and *MaMPKX-3* did not contain an activation domain, they were not used for this analysis. The remaining 22 MPK were clustered according to their phylogenetic relationship. The comparison of the complete cDNA sequences with the genome sequences revealed the intron–exon structure of the MaMPKs. All MaMPKs, in comparison to the other MPKs of the D group, had a large number of exons with variable exon lengths. In this group, the *MaMPK9-3* has 13 exons, while

MaMPK20-3 and *MaMAPK9-1* have 12 exons each. The MaMPKs from the other groups had similar number of exons with variable lengths. The number of introns ranged from 1 to 12. Only in group A, most of the genes had a similar genomic structure (Fig. 4).

Expression of MPKs during ripening in banana fruit

MPKs have been shown to be involved in the ethylene signal transduction pathway, which leads to modulation of the expression of ethylene responsive factors. Exposure of banana fruits to ethylene initiates the ripening process, leading to softening, aroma production, and a change in peel color (Lohani et al. 2004). As the role of MPKs is not known during fruit ripening, we studied the expression of identified banana MPKs during the course of ripening (from day 0 to 8). Out of all the MPKs, three MAPKs (*MaMPK20-5*, *MaMPK20-4*, and *MaMPK20-6*) did not show any ripening related response in the banana fruits and could be involved in other pathways or processes. Expression of eight MaMPKs (*MaMPK3-1*, *MaMPK11-4*, *MaMPK11-2*, *MaMPK4*, *MaMPK9-1*, *MaMPK9-2*, *MaMPK9-3*, and *MaMPK4*) was reduced postethylene treatment, suggesting that these MaMPKs might be negatively regulated by ethylene. Most of the ripening-induced MaMPK expression showed a postethylene increase

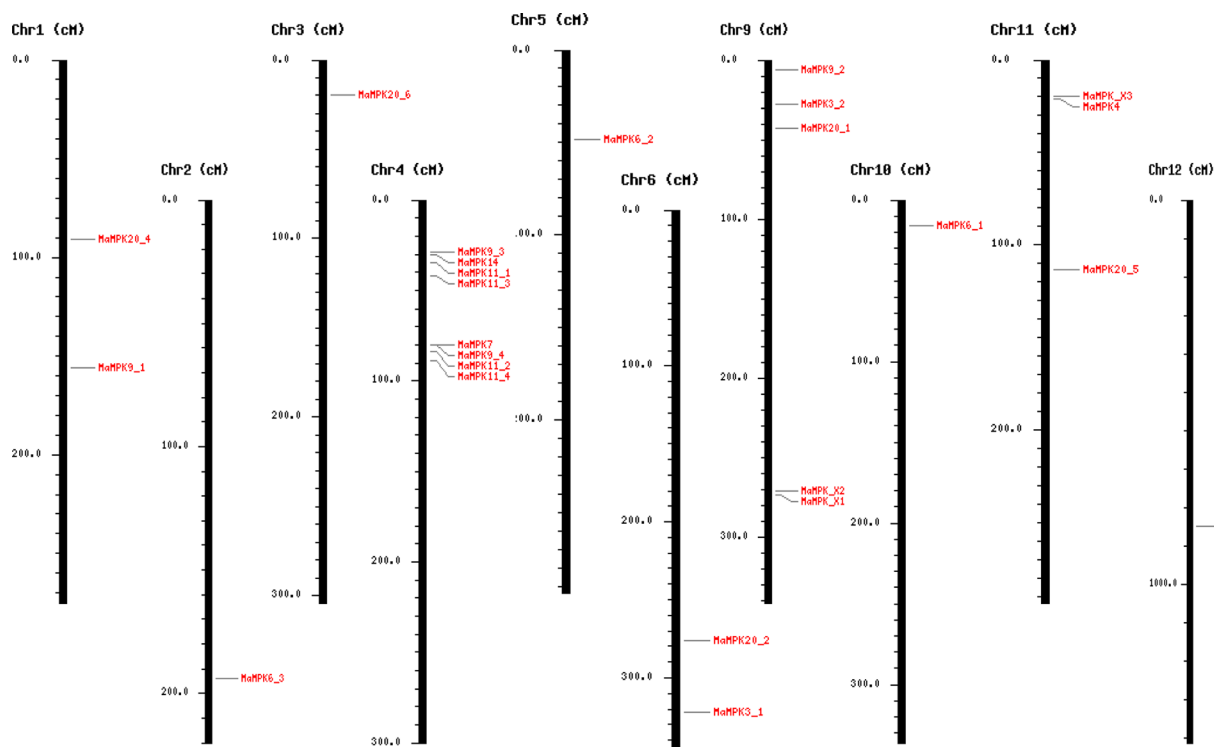


Fig. 3 Localization of the MaMPK genes on banana chromosomes. The *black lines* on the banana chromosomes (*vertical*) indicate the positions of the MPK genes

(Fig. 5). For most of the ethylene-responsive MPKs, significantly high expression on day 6 of ripening was observed. The MaMPKs that showed the highest expression at 6 days postethylene exposure generally showed a steady increase from day 2 onwards. Between this time, most of the ripening-related changes take place in fruit, followed by spoilage (Bapat et al. 2010). *MaMPK20-2* and *MaMPK20-3* showed a significant increase in their expression at 2 days postethylene exposure, indicating that these MPK are early ethylene responsive. A set of MPKs (*MaMPK20-1*, *MaMPK11-1*, *MaMPK9-4*, and *MaMPK7*) was highly expressed in the later stages of softening when the fruit has just begun senescence. It seems that these MaMPKs might play an important role in senescence or in the response to pathogen stress, as at this stage of ripening, fungal infection begins to take place. Previous studies also suggest that a set of genes related to defense and stress is induced during banana fruit ripening (Kesari et al. 2007, 2010).

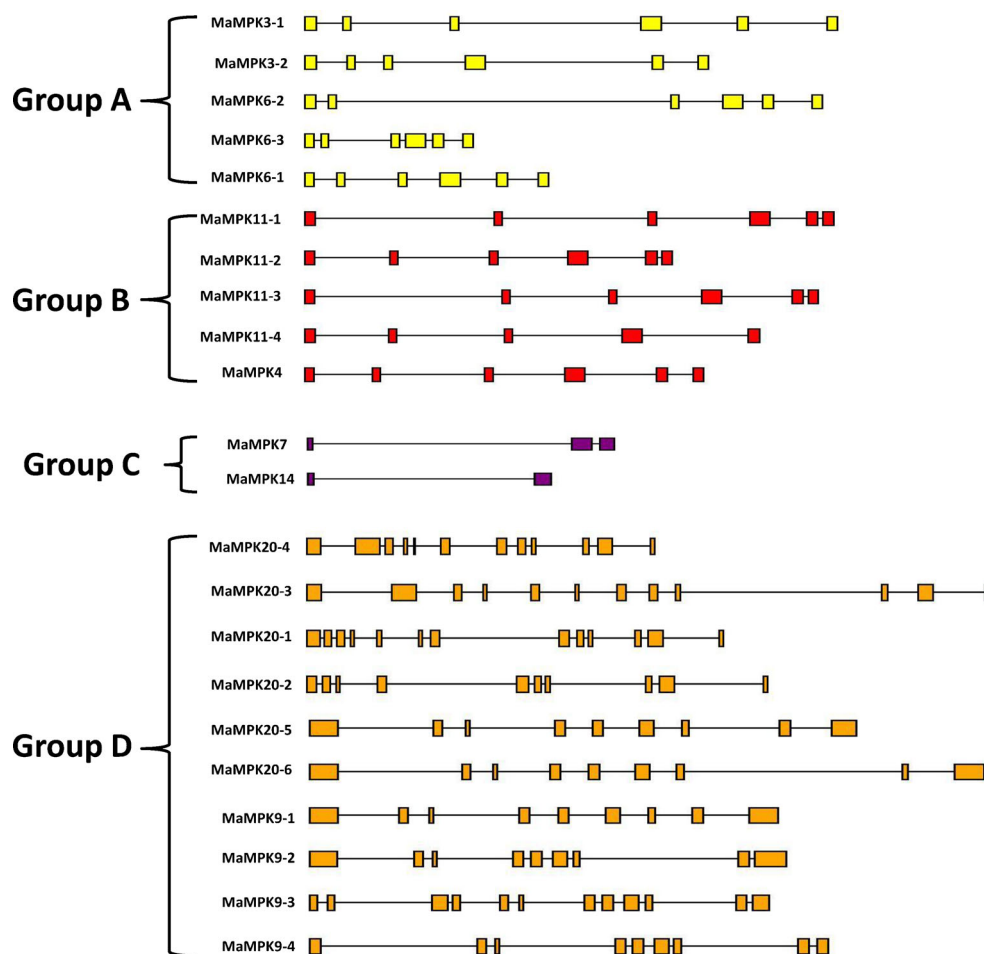
As 1-MCP is known to bind ethylene receptors, it inhibits banana fruit ripening (Lohani et al. 2004). In our study, fruit exposed to 1-MCP treatment before ethylene exposure did not ripen. Similar observations have been made previously by our group (Lohani et al. 2004) and other groups working on banana fruit ripening. It has been shown that 1-MCP treatment before ethylene exposure blocks the increase in ethylene exposure (Lohani et al. 2004; Kesari et al. 2007; Asif et al. 2009a, b; Pathak et al. 2003). To study the effect of 1-MCP treatment

before ethylene exposure on the expression of MaMPKs, we analyzed the expression in day 4 fruits treated with ethylene and 1-MCP followed by ethylene exposure (Fig. 6a). For most of the MPKs, the expression of MaMPKs in 1-MCP-treated fruits was much lower in comparison to fruits treated with only ethylene, indicating that ethylene regulated the expression of the MPKs. Surprisingly, expression of *MaMPK9-3* increased many fold in 1-MCP treated fruits in comparison to fruits treated with only ethylene. As many aspects of signaling and metabolic changes during fruit ripening are clearly associated with the peel tissue also, we compared expression of MaMPKs in peel tissue with pulp tissue in day 4 fruits treated with ethylene expression (Fig. 6b). Interestingly, expression of *MaMPK6-3*, *MaMPK20-1*, *MaMPK20-2*, and *MaMPK20-3* were significantly more in peel tissue in comparison to pulp tissue of same ripening stage. This suggests that these genes might be involved in signaling and metabolic changes in peel tissue during fruit ripening.

Expression of MaMPKs during pathogen stress

In a previous study (Li et al. 2012), the root transcriptomes of two banana cultivars, one that was susceptible to the fungus *F. oxysporum* and another that was resistant, were established. The transcriptomic data from this study was used, and the reads were mapped onto the 25 MaMPK genes identified in banana. Of the total 25 MPKs, the expression of 20 MPK

Fig. 4 Schematic diagram of the intron and exon structure of the MaMPK genes. Introns (black lines) and exons (colored boxes) of the MaMPKs are grouped according to the phylogenetic classification



genes was observed in the root transcriptome as well (Fig. 7). Out of all the MaMPKs that were present in the root, some showed differential expression between two varieties. Significant differences in the expression of *MaMPK20-3* were observed in both cultivars postfungus infection. The genes that showed a differential response to fungal attack suggest that MPKs are differentially regulated by fungal attack and may play a role during biotic stress.

Promoter sequence analysis of MaMPKs

To understand the transcriptional control of the *Musa* MPK genes, sequences of 1.5 kbp upstream region from the translation initiation codon were extracted for the 25 genes and subject to analysis on the Plant Care server. The analysis suggested that various *cis*-acting elements related to plant growth, development, and stresses are present in different MaMPKs. The Skn-1 motif that is responsible for endosperm-specific expression was present in all the MaMPK promoters except for *MaMPK6-3* (Table 2). The other motifs that were highly conserved in all the MaMPK promoters were the MeJ responsive elements,

salicylic acid responsive element, and drought-inducible responsive element. The fungal responsive element, abscisic acid responsive element, heat-shock element, and gibberellin-responsive element were present in more than 10 of the MaMPK promoters. The ethylene responsive element ERE was present in seven MaMPKs, whereas the auxin responsive element was present in six MaMPKs (Table 2).

Functional network predicted from the MPK genes

The study of potential metabolic networks associated with a gene family is a very useful method for maximizing information for the putative function of genes (Asif et al. 2009a, b; Misra et al. 2010; Dubey et al. 2010). To identify the involvement of banana MPKs in different cellular functions, the transcriptome interaction networks of the MaMPKs were analyzed using the software Pathway Studio (Ariadne Genomics, USA). The pathway diagram showed the MPK genes involved in mainly 15 cellular processes (Fig. 8). The nodes show the metabolomes that may modulate molecular functions. The major cellular processes associated with the MaMPKs are abscission, defense response, response to

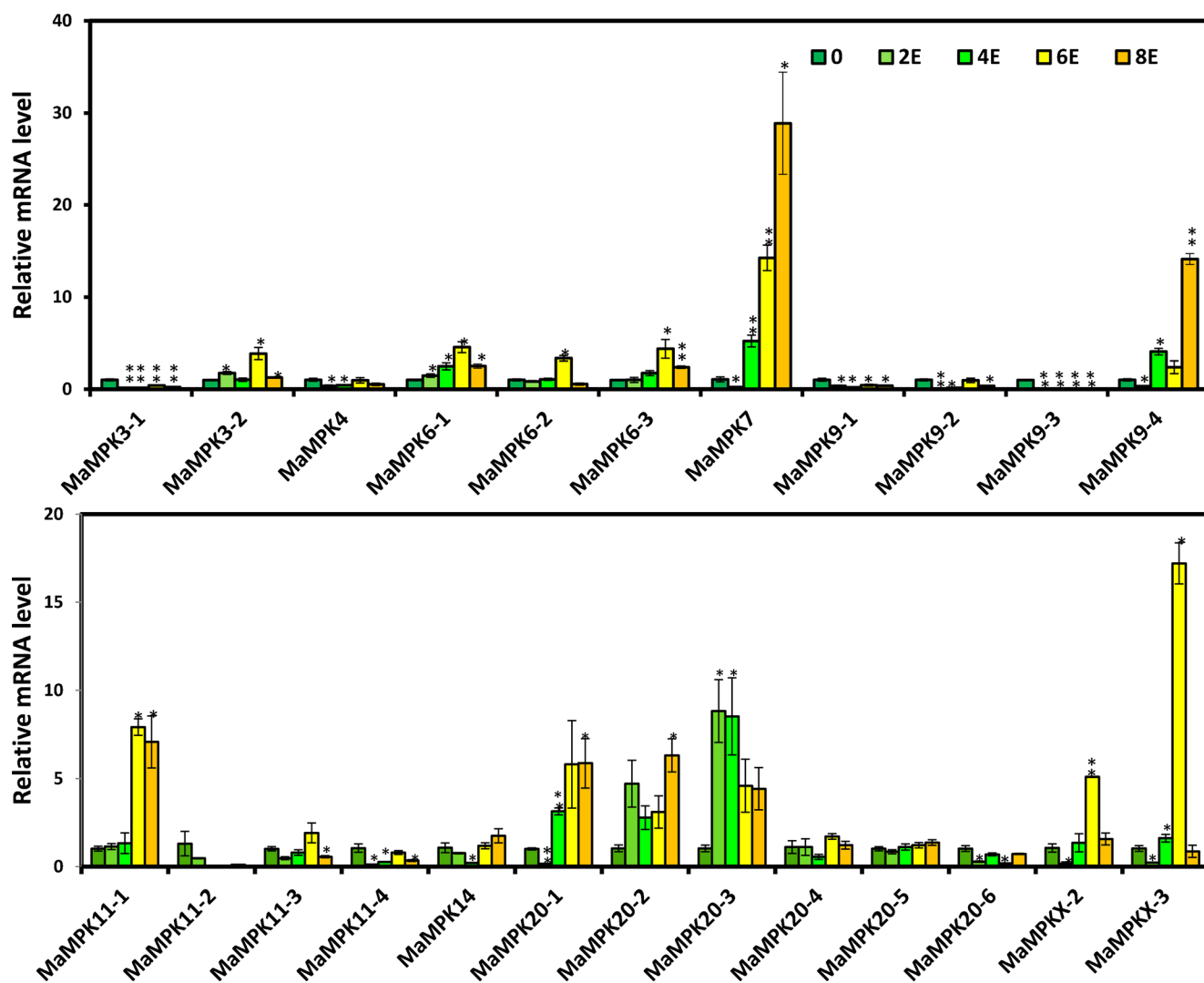


Fig. 5 The expression profiles of the members of the banana MPK family during ethylene-induced ripening. Quantitative real-time PCR of the MaMPK genes was carried out using total RNA isolated from fruit tissues. 0 to 8E represent the days postethylene treatment in the banana fruits

ethylene stimulus, immune response, plant development, and cell death. Analysis also shows involvement of these MPKs during cold, mechanical, osmotic, and drought stresses. Ripening-related genes like members of ACS, CTR, ETR, and ERF gene families were also observed in the network developed using MPKs. This suggests direct role of these in ripening process. Our promoter analysis also suggested the presence of *cis*-acting regulatory elements involved in processes identified in network. The results are indicative of a vast reservoir of information that needs to be utilized systematically to understand the detailed functions of the MaMPKs.

Discussion

MPKs are encoded by a multigene family and known to be involved in transferring signals to targets to regulate cellular

processes. So far, significant analyses have been conducted to identify and characterize the MPK gene families in several model plants, although no systematic analysis of the MPK gene family have been reported in fruit crops. In the present study, we carried out a genome-wide analysis to identify members of the MPK gene family in banana. Our analysis resulted in the identification of 25 MPK genes using the HMM profile. Furthermore, these members have been characterized on the basis of their structural diversity, phylogenetic relationships, conserved protein motifs, chromosomal location, gene duplications, exon/intron organization, promoter region, and expression profiles. Of all the members identified, three MPKs were truncated in the genome and in the coding region. The total number of MPKs in banana was more or less similar to the number of MPK genes present in other plant species, such as *Arabidopsis* (20), *Oryza* (17), *Populus* (21), tobacco (17), and tomato (16). The presence of truncated MPK genes in *M. acuminata* is a novel finding indicating

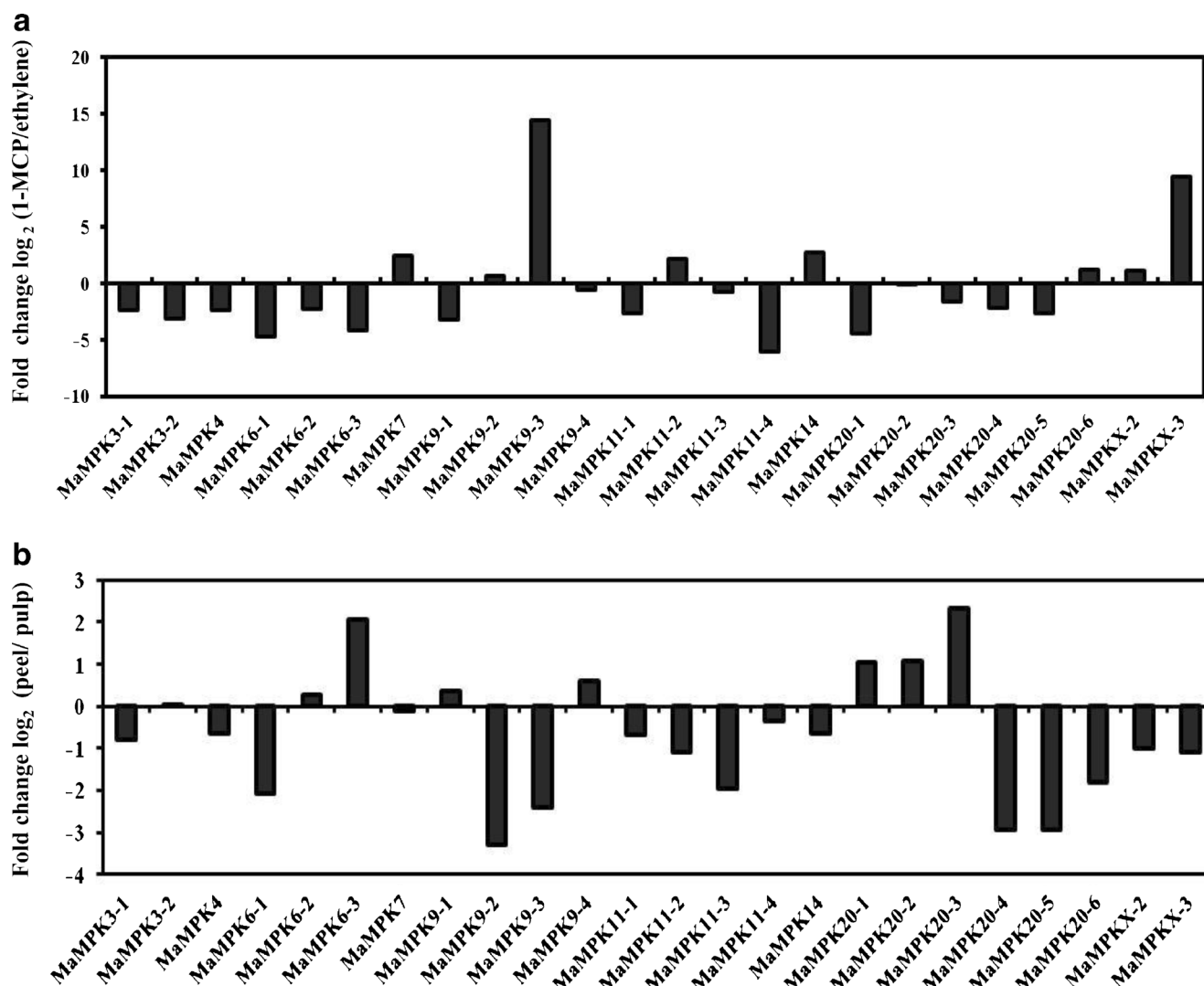


Fig. 6 Effect of 1-MCP on the ethylene modulated expression profile and in peel tissue of the members of the banana MPK genes. Quantitative real time PCR of the MaMPK genes was carried out using total RNA isolated

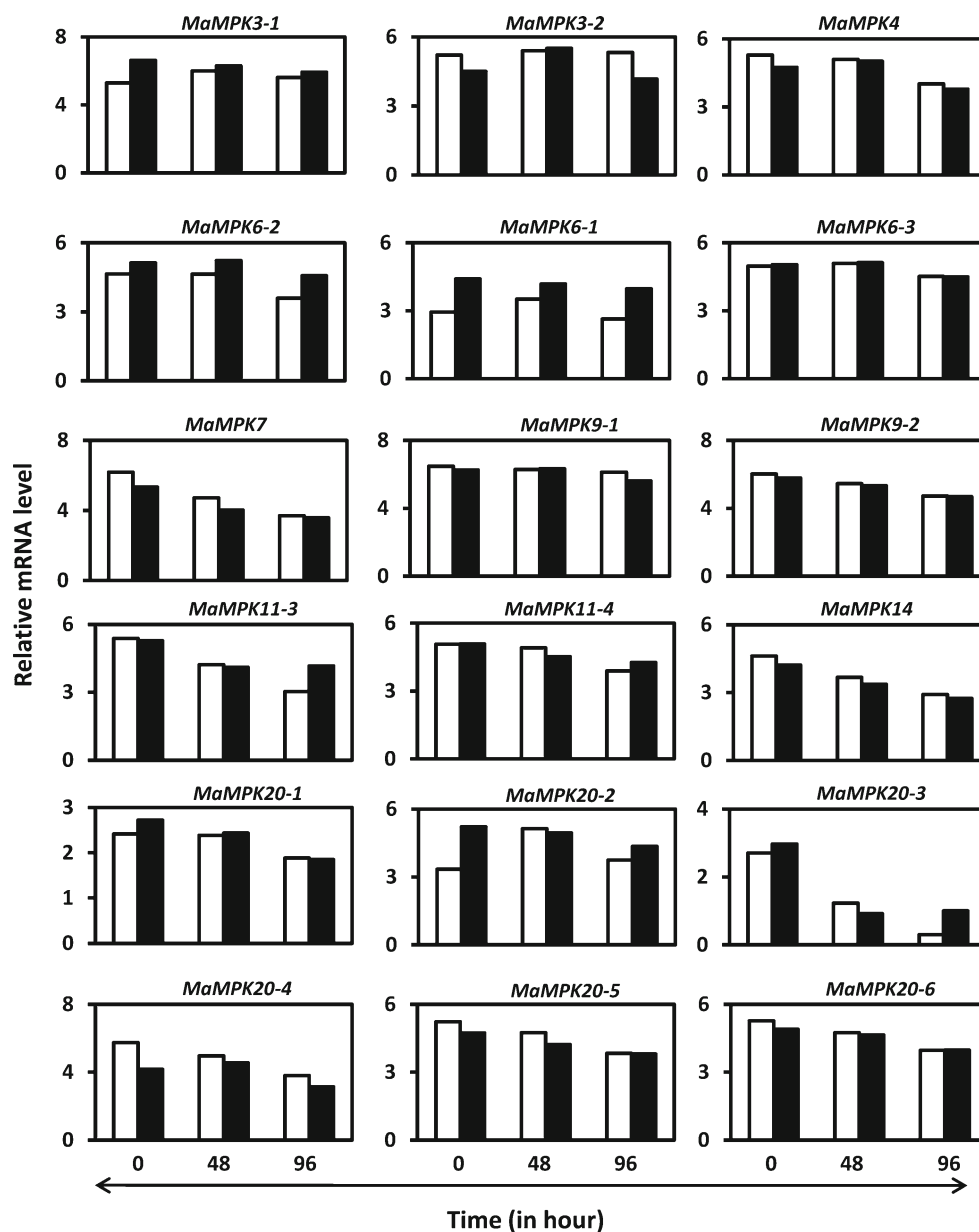
from the pulp and peel tissues of 4 days after ethylene and 1-MCP exposure to banana fruit. Fold change was calculated to study relative expression between ethylene and 1-MCP treatments (a) and peel and pulp tissues (b)

loss of function, and it may be due to rearrangements in the banana genome. The *M. acuminata* MPK genes clustered in the four groups (A–D), similarly to other plants (Fig. 2). The genes with a TEY domain were classified in groups A–C; however, the MaMPKs present in group C were shorter both at the N and C termini. This was the first instance of MPKs that were shorter at both the ends, but contained the activation loop. None of the MPKs in banana had the MEY domain that is present in one MPK in rice, tomato, tobacco, and *Populus* (Rao et al. 2010). The catalytic domain present in the TEY MPKs is conserved in MaMPKs. Similar to other species, group C contains a mutated version of the catalytic domain. The catalytic domain was not present in the members of D group. There were 10 MPKs in group D in banana, and all these members had the extended C terminus region that is present in MPKs from other plant species (Kong et al. 2012). The XI domain, which is characteristic of MPKs from other

species, is also present in the banana MPKs. Many of the MPK in banana are paralogs but not duplons.

The structural analysis of the MaMPK genes showed that the intron–exon structure was similar to that of other plant species (Chen et al. 2012; Kong et al. 2012). All the members of groups A and B had close to six exons as observed in *Solanum* and other species. The members of group C have less than four exons in all the species studied, including banana. Similar to other species, the members of group D have up to 12 introns and exon. The similar gene structures of the different groups across species suggest that these genes have evolved in a similar fashion in all the species, and each group may have evolved from a single origin. The phylogenetic and structural analysis also reveals that the evolution of these genes may have originated from one type of species, and the appearance of paralogs was a

Fig. 7 The expression profiles of the members of the banana MPK family during fungus infection in the banana root. Raw reads of the transcriptome data from fungus-resistant (*white bar*) and fungus-susceptible (*black bar*) Cavendish banana roots were used for calculating the expression levels



later event. It is interesting to note that many members of the MPK genes have paralogs, but none of them are duplcons, indicating that these paralogs have evolved over the time for new functions or they slowly took over the functions of old genes that may have been disrupted by mutations or truncations.

The expression of the MaMPK genes was studied during the entire phase of ripening. In addition, expression of the MaMPK genes was deduced during fungal infection using available transcriptome data. At least one MPK from each group showed significantly high expression during the ripening of the banana fruit. One of the highly upregulated genes was *MaMPK11*, which is present in group B. The MPKs from this group are known to be involved in negative regulation of systemic acquired resistance, which could also be the reason

for the high expression of *MaMPK11-1* in the later stages of ripening and senescence of the fruit. These MPKs may be more involved in the wounding and pathogen response than the ripening response. *MaMPK6-1*, *MaMPK6-3*, *MaMPK20-1*, *MaMPK20-2*, and *MaMPK20-3* could be ripening-related MPKs because they showed a steady increase in expression as ripening progressed (Fig. 5). During the ripening of the banana fruit, the *MaMPK20-5* and *MaMPK20-6* did not show any significant modulation in expression as the ripening progressed. The expression of all these MaMPKs was reduced in the presence of 1-MCP, a potent inhibitor of ethylene perception, indicating that these genes may be directly regulated by ethylene (Fig. 6a). Of these genes, *MaMPK6-1* and *MaMPK6-3* belong to group A, and *MaMPK20-1*, *MaMPK20-2*, and *MaMAPK20-3* belongs to group D. The

Table 2 *cis*-Acting regulatory elements present in promoters of various MAP kinase genes from banana

Gene name	Motifs related to growth and development	Motifs related to stress response
<i>MaMPK3-1</i>	CCGTCC-box, circadian, GCN4_motif, O2-site, Skn-1_motif	ARE, Box-W1, MBS, MBS, P-Box, TC-rich repeats
<i>MaMPK3-2</i>	CAT-box, CCGTCC-box, circadian, O2-site, Skn-1_motif	ARE, Box-W1, CGTCA-motif, LTR, MBS, MBS, P-Box
<i>MaMPK4</i>	CCGTCC-box, circadian, Skn-1_motif	ARE, CGTCA-motif, GARE-motif, MBS, MBS
<i>MaMPK6-1</i>	CAT-box, circadian, GCN4_motif, Skn-1_motif	ABRE, Box-W1, CGTCA-motif, MBS, MBS
<i>MaMPK6-2</i>	CCGTCC-box, circadian, O2-site, Skn-1_motif	ABRE, ARE, Box-W1, CGTCA-motif, GARE-motif, MBS, MBS, TC-rich repeats
<i>MaMPK6-3</i>	CAT-box, circadian	ARE, CGTCA-motif, GARE-motif, HSE, HSE, LTR, MBS, MBS, TC-rich repeats
<i>MaMPK7</i>	GCN4_motif, O2-site, Skn-1_motif	ARE, AuxRR-core, ERE, HSE, HSE, MBS, MBS
<i>MaMPK9-1</i>	CCGTCC-box, circadian, GCN4_motif, Skn-1_motif	GARE-motif, MBS, MBS, P-Box
<i>MaMPK9-2</i>	circadian, Skn-1_motif	TC-rich repeats
<i>MaMPK9-3</i>	CAT-box, circadian, O2-site, Skn-1_motif	ARE, CGTCA-motif, ERE, HSE, HSE, MBS, MBS, TC-rich repeats
<i>MaMPK9-4</i>	circadian, GCN4_motif, O2-site, Skn-1_motif	ARE, Box-W1, ERE, LTR, MBS, MBS, WUN-motif
<i>MaMPK11-2</i>	circadian, GCN4_motif, O2-site, Skn-1_motif	ABRE, ARE, Box-W1, CGTCA-motif, GARE-motif, MBS, MBS, TC-rich repeats
<i>MaMPK11-3</i>	circadian, Skn-1_motif	ABRE, ARE, CGTCA-motif, MBS, MBS
<i>MaMPK11-4</i>	CCGTCC-box, circadian, Skn-1_motif	ABRE, Box-W1, CGTCA-motif, GARE-motif, MBS, MBS, TC-rich repeats, WUN-motif
<i>MaMPK14</i>	circadian, GCN4_motif, Skn-1_motif	ABRE, AuxRR-core, CGTCA-motif, MBS, MBS
<i>MaMPK20-1</i>	CAT-box, circadian, Skn-1_motif	Box-W1, CGTCA-motif, MBS, MBS, P-Box, TC-rich repeats
<i>MaMPK20-2</i>	circadian, GCN4_motif, O2-site, Skn-1_motif	ABRE, ARE, CGTCA, GARE, HSE, HSE, MBS, MBS, TC-rich repeats, WUN-motif
<i>MaMPK20-3</i>	circadian, GCN4_motif, Skn-1_motif	CGTCA-motif, GARE-motif, MBS, MBS, TC-rich repeats, WUN-motif
<i>MaMPK20-4</i>	CAT-box, circadian, Skn-1_motif	ABRE, ARE, Box-W1, ERE, GARE-motif, TC-rich repeats
<i>MaMPK20-5</i>	CAT-box, circadian, Skn-1_motif	ABRE, ARE, Box-W1, HSE, HSE, LTR, MBS, MBS, TC-rich repeats
<i>MaMPK20-6</i>	circadian, GCN4_motif, Skn-1_motif	ARE, Box-W1, CGTCA-motif, GARE-motif, HSE, HSE, MBS, MBS, WUN-motif
<i>MaMPKX-1</i>	circadian, O2-site, Skn-1_motif	ABRE, AuxRR-core, CGTCA-motif, GARE-motif, MBS, MBS
<i>MaMPKX-2</i>	circadian, GCN4_motif, O2-site, Skn-1_motif	ABRE, AuxRR-core, CGTCA-motif, ERE, HSE, HSE, MBS, MBS, TC-rich repeats
<i>MaMPKX-3</i>	Skn-1_motif	ABRE, Box-W1, CGTCA-motif, ERE, HSE, HSE, MBS, MBS

group A genes have been shown to be involved in the response to biotic and abiotic factors, and both *MaMPK6-1* and *MaMPK6-3* showed the highest expression at day 6 postethylene exposure. *MaMPK6-2*, which also belongs to this group, also showed a peak at day 6 postethylene exposure. This is the stage where major cell wall breakdown for softening occurs, which is similar to the wounding response. Apart from MPK, several genes involved in the biotic and abiotic stress responses have been shown to be induced during banana fruit ripening (Kesari et al. 2007, 2010). The genes from this group in other plants include WIPK and SIPK (AtMPK3/6, OsMPK6), thus indicating the involvement of these genes in ethylene-regulated softening of the banana fruit. The other significantly regulated ethylene response genes were *MaMPK20-2* and *MaMPK20-3*, which showed significantly high expression just after ethylene treatment, indicating the role for these genes in the ethylene signaling pathway. The role of the MPK

cascade has been shown to be involved in the ethylene signaling pathway (Bapat et al. 2010). Both these genes are group D MPKs. Interestingly, this group is the least understood group in the MPK phylogeny. *MaMPK20-1* also belongs to this group, and it showed a significantly high expression from 4 days onwards. It has been previously reported that, in banana, a biphasic pattern of ethylene evolution during ripening is observed (Pathak et al. 2003), one at day 2, and the other at day 4. This *MaMPK20-1* may be involved in the later stage of ethylene production and signaling. Similar to pulp tissue, expression of MaMPKs was also modulated in peel tissue. However, this modulation was less in comparison to pulp tissue (Fig. 6b) except for *MaMPK6-3*, *MaMPK20-1*, *MaMPK20-2*, and *MaMPK20-3*, which showed significantly higher expression in peel tissue of same ripening stage. As peel tissue of banana also undergoes several changes during ripening, these MaMPKs might

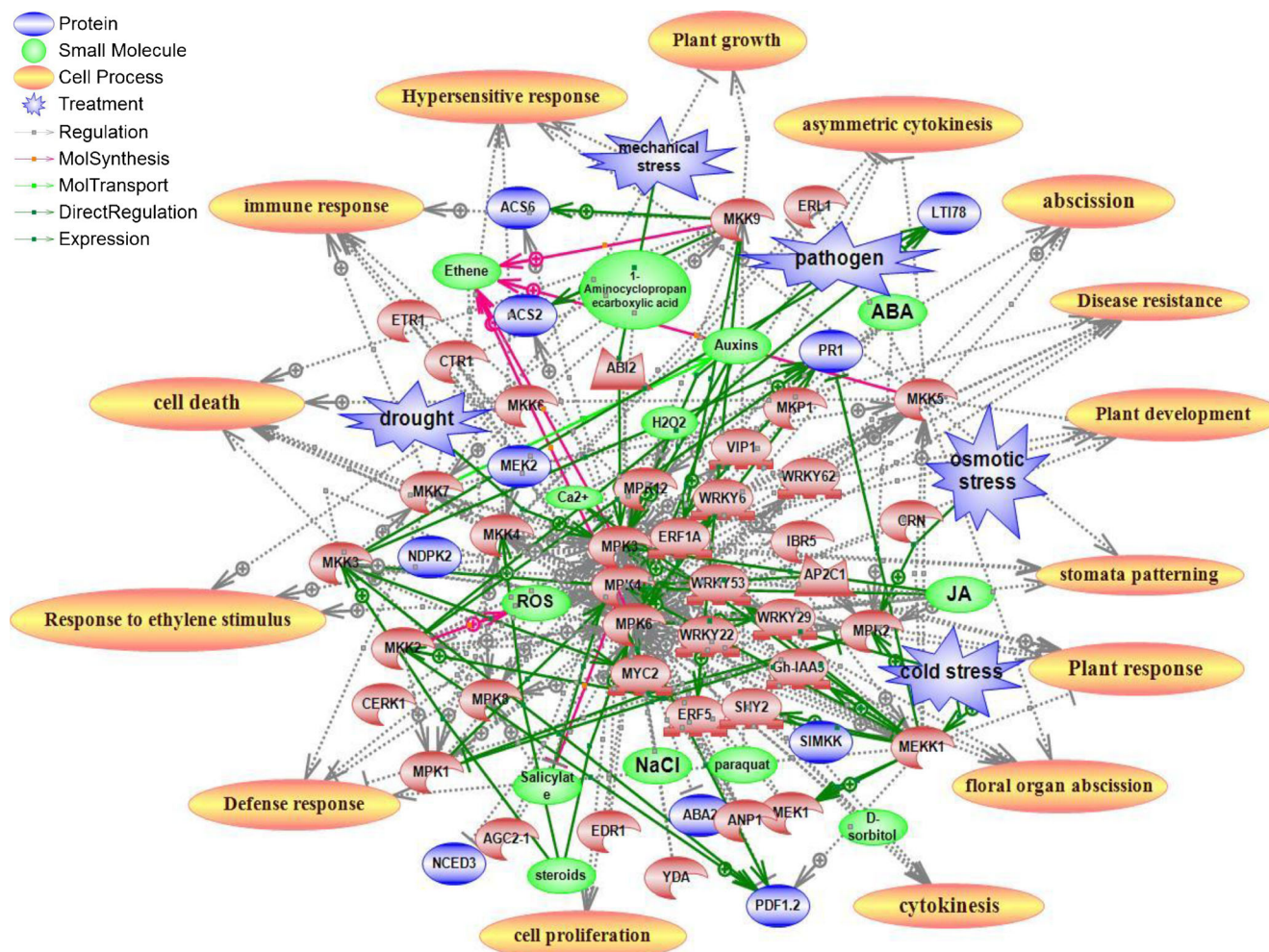


Fig. 8 Involvement of the banana MPK gene family in different cellular processes. The MaMPK genes were analyzed for their cellular pathways and interactions using the program Pathway Studio. The pathway diagram shows that the MPK genes are involved in different cellular

processes including abscission, defense response, response to ethylene stimulus, immune response, plant development, and cell death. The nodes show the metabolomes that may modulate the molecular functions

be involved in regulation of peel-specific processes during ripening. The further analysis of members of this group in relation to ethylene biosynthesis and signaling may provide important information for their role in these pathways.

The other significantly highly expressing genes were *MaMPK9-4*, *MaMPK7*, and *MaMPKX-2*, which showed high expression in the later stages of ripening, with a significant increase after day 4 postethylene exposures. Both the *MaMPK9-4* and *MaMPK7* showed a peak at day 8 and *MaMPK-X2* at day 4 postethylene exposure. These genes collectively belong to group C, which contains well-characterized genes involved in wounding, JA, SA, ABA, H_2O_2 , salt stress, etc. The promoter region of these MPKs contains *cis*-elements that are known to be involved in these stresses (Table 2).

The study of the promoter regions of the MaMPK genes is important to understand their functional diversity. It was

interesting to note that the Skn-1 motif was present in nearly all the *Musa* MPK promoters except for *MaMPK6-3* (Table 2). This motif was shown to be involved in endosperm expression, indicating that most of these MPK were able to express efficiently in the fruit tissues. Similarly, the Skn-1 motif was also predominantly present in the *Solanum* MPK. The other highly represented motif was the MBS motif, which is involved in the drought response. This motif is present in 22 of the 25 banana MPKs. MPKs have been known to be involved in defense against reactive oxygen species (ROS) and biotic and abiotic stresses. As drought is an important form abiotic stress, the plant MPK genes may have evolved to cope up with this type of stress. A large number of the MPKs contain hormone-related motifs, especially for MeJ, salicylic acid, abscisic acid, and gibberellin responses. These hormones, particularly salicylic acid, MeJ, and abscisic acid, have been shown to be involved in the ROS response. The group D MPKs has been shown to be highly responsive to ROS.

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